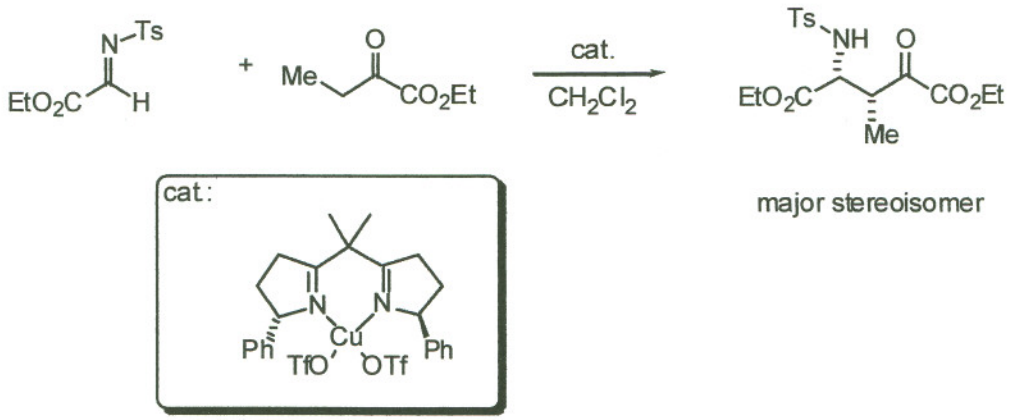
Organic Cume
Jan 15, 2004

1. Show the detailed mechanism of the acid-catalyzed Mannich reaction between ammonia, formaldehyde, and acetone as the reagents. Show electron pairs, correct 'arrow pushing' to indicate bond formation and cleavage.

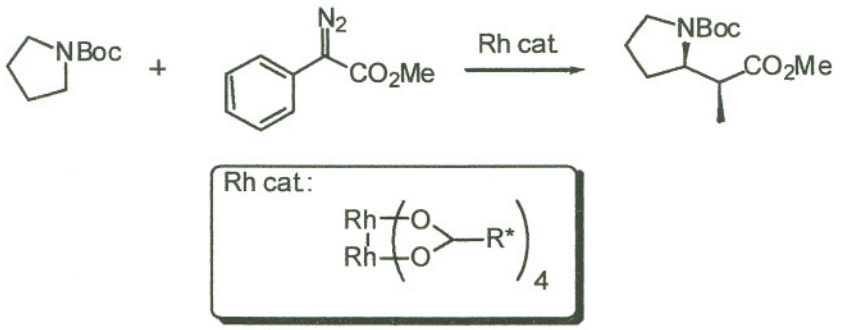
2. Same as question 1, except with base as the catalyst.

3. Show the product(s) of the Mannich reaction of benzaldehyde, ammonia, and propiophenone ($C_6H_5COC_2H_5$). Draw all possible stereoisomers, label the stereocenters as *R* or *S*.

4. Suggest a transition state assembly that accounts for both the relative and absolute configuration of the following reaction. Indicate whether the topicity of the reaction is *lk* or *ul*.



5. Last semester, Professor Huw Davies showed us some of his results on catalytic asymmetric C–H activation as a surrogate for the Mannich reaction. For example:



Without regard to stereochemistry, provide a mechanism of this reaction, and explain how (or why) it is called a 'surrogate for the Mannich reaction.